Summary in Graph

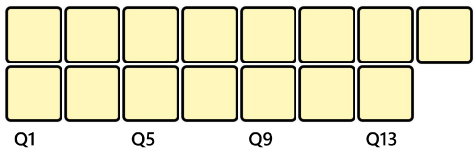
Exam Summary (GO Classes DA Test Series 2026 | Probability and Statistics | Hypothesis Testing | Topic Wise Test 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (43) out of 427 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The Cashier of SM Supermarket claims that the mean Monthly salary of their employees is Php$22,500.00$ with a standard deviation of Php$7,000.00$ A researcher takes a random sample of 80 employees and found out that they have a mean monthly salary of Php$18,750.00$. Do the 80 selected employees have lower salaries than the rest ? Test the claim at 5% level of significance. The appropriate $H_0$and H_1: are:

A. $\mathrm{H}_0 : \mu = \mathrm{P}22,500;\ \mathrm{H}_1 : \mu < \mathrm{P}22,500$

B. $\mathrm{H}_0 : \mu \leq \mathrm{P}22,500.\ ;\mathrm{H}_1 : \mu > \mathrm{P}22,500$

C. $\mathrm{H}_0 : \mu \geq \mathrm{P}22,500;\ \mathrm{H}_1 : \mu < \mathrm{P}22,500$

D. $\mathrm{H}_0 : \mu\mathrm{E} = \mathrm{P}22,500;\ \mathrm{H}_1 : \mu \neq \mathrm{P}22,500$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 03sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Type II error in hypothesis testing is

A. acceptance of the null hypothesis when it is false and should be rejected
B. rejection of the null hypothesis when it is true and should be accepted
C. rejection of the null hypothesis when it is false and should be rejected
D. acceptance of the null hypothesis when it is true and should be accepted

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following statements are true?

A. Z-distribution is symmetric about mean.
B. T-distribution is symmetric about mean.
C. Chi-squared distribution is symmetric about mean.
D. Z-distribution has mean $= 0$, variance $= 1$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Probability

In a class test the mean test score for the entire class is 70 with standard deviation equal to 10 . The mean score of a random sample of 64 female students is 73 . The z-value is?

Your Answer: | Correct Answer: 2.4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Statement 1: The acceptance region of $\alpha = 0.1$ is more than the acceptance region of $\alpha = 0.05$
Statement 2: If α (alpha) increases then rejection region increases.

A. Statement 1 is true.
B. Statement 2 is true.
C. Both statements are true.
D. Both statements are false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Assume that the hardness of iron bars produced by one plant is normally distributed with mean μ and variance σ^2. A random sample of $n = 7$ bars from the output of the plant is measured to determine whether the average hardness meets the desired figure $\mu_0 = 172$.
The following measurements:

$$167, 174, 179, 164, 163, 160, 168$$

were obtained. Test $H_0 : \mu = 172$ versus the alternative $H_a : \mu \neq 172$ using an appropriate test with $\alpha = .01$.
Which test is performed here?

A. Two tail t-test can be performed
B. Right-tail z-test can be performed
C. Left-tail z-test can be performed
D. Two tail z-test can be performed

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

In this problem, let S represent the proportion of women who agree that men are basically selfish and egotistical.
Suppose that surveys done 20 years ago showed that S was 0.40 at that time, but a new survey finds that $Y = 1260$ out of a random sample of $n = 3000$ women said that men are basically selfish and egotistical. Is there sufficient evidence to say the proportion S has increased? Define the $\mathrm{H_0}, \mathrm{H_a}$ for this problem statement.

A. $\mathrm{H_0 : S = 0.4, H_a : S < 0.4.}$

B. $\mathrm{H_0 : S = 0.4, H_a : S \neq 0.4}$

C. $\mathrm{H_0 : S = 0.4, H_a : S > 0.4.}$

D. $\mathrm{H_0 : S > 0.4, H_a : S < 0.4.}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Which of the following is the p-value for two tail test? phi represents CDF of standard normal distribution and v is the test statistic.

A. 1 - phi(v)
B. phi(v)
C. $2(1 - \mathrm{phi(v)})$
D. phi(v)-1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Which of the following is/are correct?

A. Let $X_i \sim Binomial(n, p)$, then $X_1 + X_2 + \ldots + X_n$ follows normal distribution (assume n to be large)
B. Chi-squared test can be used to test independence of two features.
C. Chi-squared test can be used for goodness of fit testing
D. As alpha increases it will become easy to reject H_0 .

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Statement 1 : P-value is the smallest level of significance that would lead to rejection of H_0.
Statement 2 : P -Value is the probability that the H_0 is false.

A. Statement 1 and 2 are true
B. Statement 1 is true
C. Statement 2 is true
D. Both statements are false.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose the jewelry of exams has a population that is normally distributed with a standard deviation of 5 . You are walking down the street and sample 9 exams from this population and obtain a mean jewelry of 28.95 and a standard deviation of 6.3802 .
Using an alpha value of $\alpha = 0.01$, is this observed mean significantly different than an expected jewelry of 27 ?
$z_{\text{critical}} = \pm 2.58$, For $\alpha = 0.01$

A. z-value $= 2.34$, The means are completely different

B. z-value $= -2.34$, The means are same

C. z-value $= -1.17$, The means are completely different.

D. z-value $= 1.17$, The means are the same.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose the courage of psychologists has a population that is normally distributed with a standard deviation of 10 . You decide to sample 57 psychologists from this population and obtain a mean courage of 34.81 and a standard deviation of 9.0579.
Using an alpha value of $\alpha = 0.05$, is this observed mean significantly greater than an expected courage of 34 ?
$Z_{critical} = 1.64$, for $\alpha = 0.05$

A. z-value observed is 0.61 , we need to perform two tailed test and the observed mean and population mean are the same.

B. z-value observed is 0.61 , we need to perform a right tailed test and the observed mean and population mean are the same.

C. z-value observed is 0.61 , we need to perform a left tailed test and the observed mean and population mean are the same.

D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Suppose the courage of hateful republicans has a population that is normally distributed with a standard deviation of 6 . Let's pretend that you sample 58 hateful republicans from this population and obtain a mean courage of 23.59 and a standard deviation of 6.61.
Using an alpha value of $\alpha = 0.05$, is this observed mean significantly less than an expected courage of 26 ?
$\mathrm{Z\text{-}value}$ for $\alpha = 0.05$ is 1.64

A. The courage of hateful republicans is significantly less than 26.

B. The courage of hateful republicans is significantly more than 26.

C. we perform two tailed test, z-value is $+3.06$

D. we perform left tailed test, z-value is -3.06

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following is/are true?

A. We can use p-value and significance level to reject H_0, If p-value is less than alpha we reject H_0.
B. We can use p-value and significance level to reject H_0 , If p-value is greater than alpha we reject H_0.
C. The Z-test is performed when the sample size is large and the variance of the population is known.
D. T-test is performed when sample size is less and population variance is not known.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider covid-19 situation, you are the doctor(imagine). When the test is positive we give treatment, else we don't. Assume the test can make errors, null hypothesis as the patient has covid 19.

A. More importance must be given to type 1 error than type 2 error.

B. More importance must be given to type 2 error than type 1 error.

C. Type 1 error is the probability (rejecting H_0|patient actually has covid 19).

D. Type 2 error is the probability (accepting H_0|patient actually doesn't have covid 19).

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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